Loops and trace tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Teach state changes before relying on a final number.

Supply complete loops, test zero/one/many visits and assess a fresh range trace after exposure checks.

Keep one habit: Ask what exists before the loop, which values are visited and when output occurs.

MAIN LESSON · SECTION 1

Prepare the connected bridge lesson

> Check prior exposure and keep execution practice distinct from paper reasoning.

Goal and main plan

Primary target: Trace variable values before and after each **loop iteration**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval and prerequisite guidance

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate numeric expressions using Python's grouping rules

Reference example, not the only acceptable wording: 3 * 8 + 2 or (3 * 8) + 2 gives 26. 3 * (8 + 2) gives 30 and adds two to each of three packs, not two once.

Teach for **loop**s with finite integer ranges before adding more **loop** forms. Keep range values, body executions and **output** events separate. Reference: https://docs.python.org/3/tutorial/controlflow.html#the-range-function.

The examples use a positive default step and do not require lists or while **loop**s.

Materials and readiness

Python 3 editor and **loop**s.py

Counters or numbered paper cards

Blank **trace table** with turn total before and total after columns

It reads the old 2, adds 3 and stores 5. Assignment updates a value; it is not an algebraic equality.

End goal

Trace **loop**s and **trace table**s and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Outcomes

Trace variable values before and after each **loop iteration**

Use the excluded stop value in range correctly

Explain **initialization** and test zero one and several **iteration**s

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach accumulator state and output events separately

> Keep the warm-up, picture, W2 and selected ranges explicitly labelled.

Read the task and loop-body syntax

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records state; it does not create extra instructions.

Inspect zero/one/many ranges

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

Do not rely on a variable first assigned inside a **loop**: a zero-**iteration** run might never assign it.

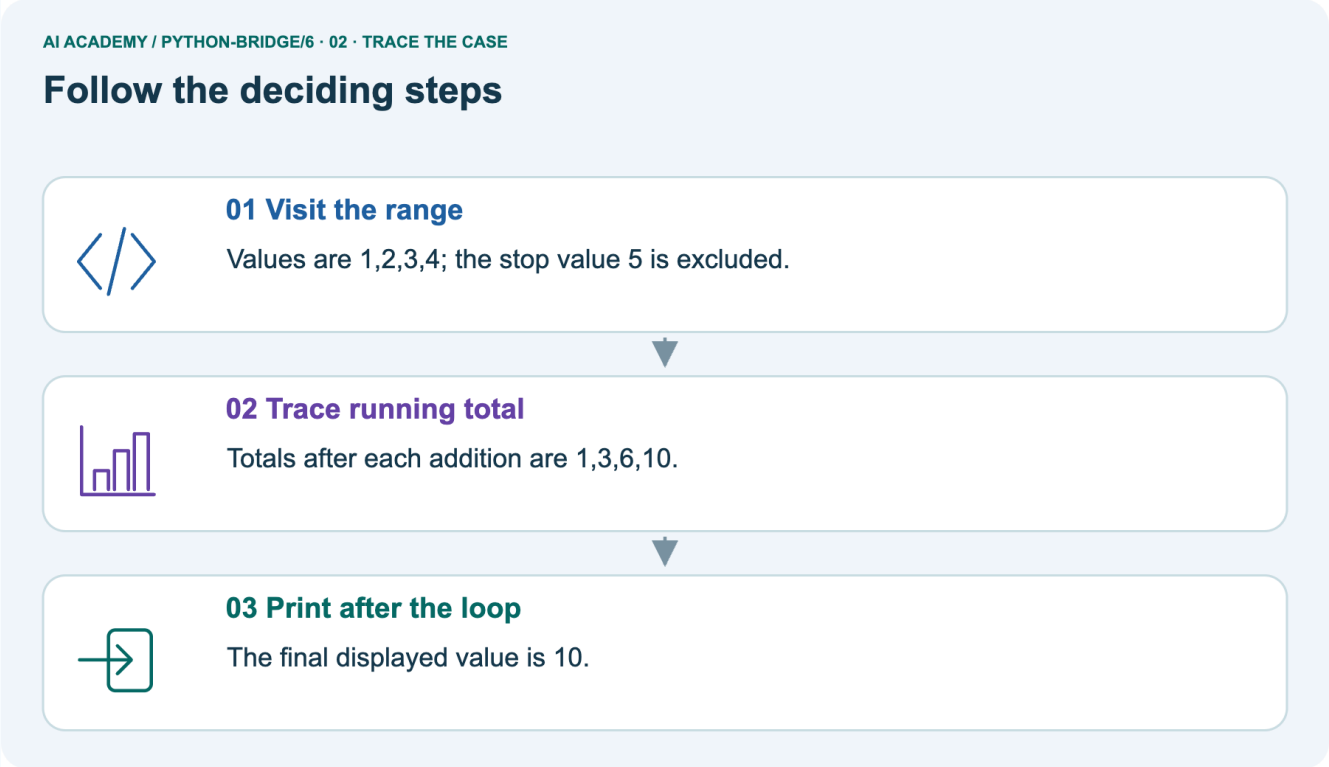
Read the pictured scope

Accumulate a short range. A running total adds the integers visited by range(1,5).

Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

The warm-up range(1,4) visits 1,2,3 and totals 6. The pictured range(1,5) visits 1,2,3,4 and totals 10. W2 uses range(2,5), which visits 2,3,4 and totals 9. Keep these three records labelled instead of swapping final numbers. The selected independent range is different again. All supplied ranges use the default positive integer step; this lesson does not implement arbitrary range-input validation.

Trace the labelled steps and original state graphic



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/6

Running total after each visited value

	Current value	Running total
Visit 1	1	1
Visit 2	2	3
Visit 3	3	6
Visit 4	4	10

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Visit the range	Values are 1,2,3,4; the stop value 5 is excluded.
2. Trace running total	Totals after each addition are 1,3,6,10.
3. Print after the loop	The final displayed value is 10.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

Match full code to before/after totals

```
total = 0
for value in range(1, 5):
    total = total + value
print(total)
```

Visited value	Total before	Total after	Output yet?
1	0	1	No
2	1	3	No
3	3	6	No
4	6	10	No

Visited value	Total before	Total after	Output yet?
After loop	10	10	Print 10 once

Predict the empty range

Change: Replace range(1,5) with range(1,1).

Prediction: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

State bounded-loop limits

This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional guided key

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Loop	A block of instructions that can run repeatedly
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result

Term	Meaning
Initialization	Giving a variable its starting value
Trace table	A record of variable values as instructions execute

State after each addition

TURN	TOTAL BEFORE	TOTAL AFTER
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

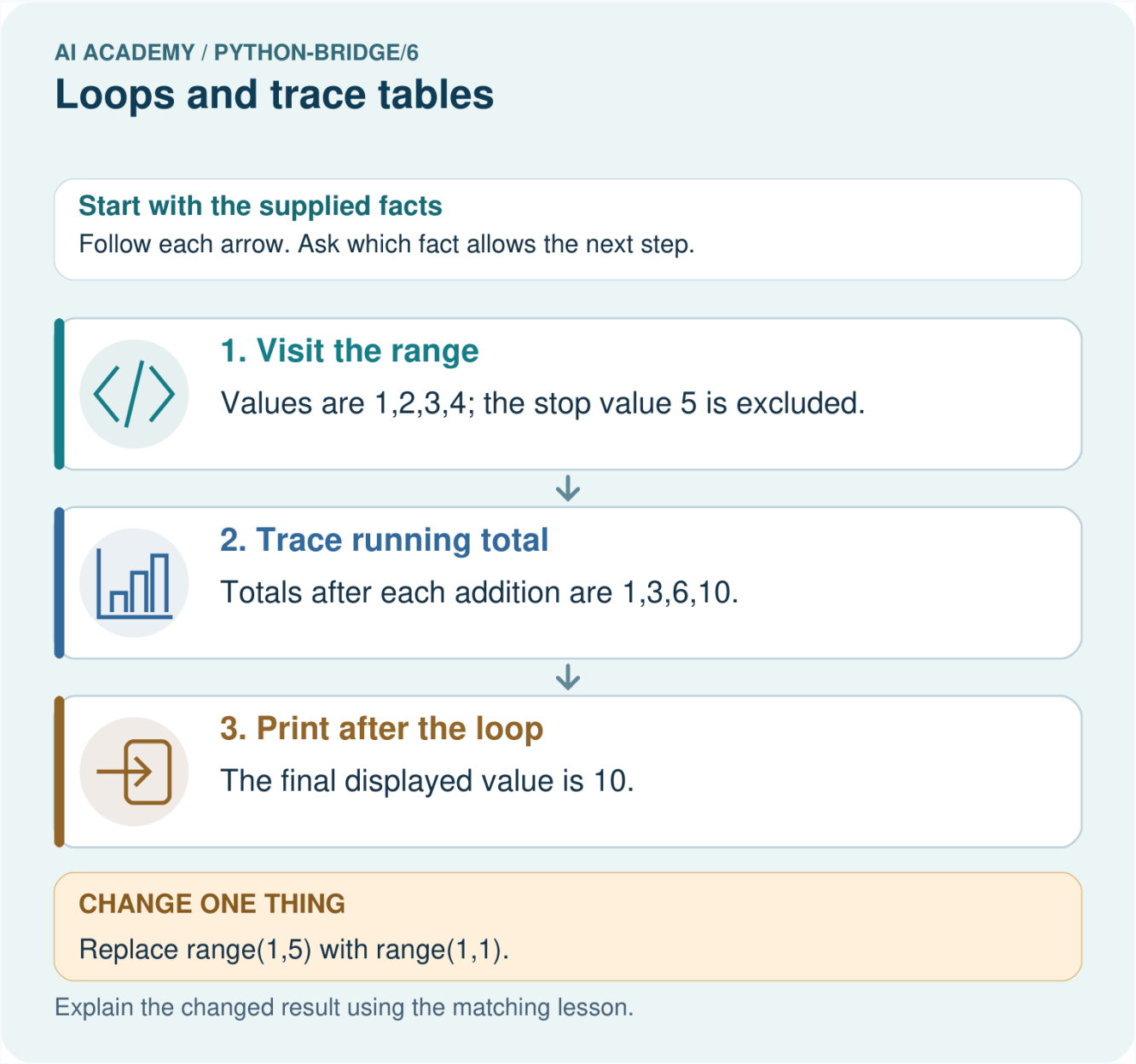
> **Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Loops and trace tables

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the trace, then fade the help

> Collect predictions before execution or feedback.

Main completion task and key

Completion task: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Teacher check: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case key

List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

The visits are 1,2,3,4 and totals are 1,3,6,10. The stop is exclusive. An empty range leaves total at its initial 0.

Changed-case reasoning: No values are visited and the final **output** is 0. The **accumulator** starts at zero and an empty **loop** never changes it.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

▼ Support and alternative plan

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve assignment using the prerequisite.
7 to 20 minutes	Model range with numbered cards and complete W1.
20 to 33 minutes	Trace each row of the worked addition and predict output.
33 to 45 minutes	Complete W2 and compare predicted and observed values.
Session 2 start to 8 minutes	Retrieve included start and excluded stop.
8 to 20 minutes	Trace the reset bug, complete W3 and discuss the guided response.
20 to 35 minutes	Collect independent W4 code and zero one and several iteration traces.
35 to 45 minutes	Use W5 and the exit to examine assumptions and revealing tests.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

▼ **Earlier wording, alternatives and picture transcripts**

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Preserve expected, observed and paper-trace records.

Supplied case materials and checked guidance

W2 rows turn 2/before 0/after 2; turn 3/before 2/after 5; turn 4/before 5/after 9. Empty correct loop prints 0, one iteration 1, several 1+2+3=6. Faulty reset final 3; inside print correct loop gives 1,3,6. Picture four-value sum 10, empty 0. Selected range(2,6) visits 2,3,4,5 and totals 2,5,9,14; range(2,2) empty leaves 0. Retain observed errors and exact prior exposure separately.

Diagnose misconceptions

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every **iteration**: Resetting the **accumulator** loses earlier work.

An unindented print repeats with the body: Indentation determines which statements repeat.

Zero **iteration**s mean the entire program does nothing: **Initialization** and statements after the **loop** still execute.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected bridge transfer task

> If a learner has seen this exact case, use a fresh parallel input and record exposure/support.

Selected independent task and checked key

Selected case: List every value visited and each running total. Then predict the result for range(2,2).

```
total = 0
for value in range(2, 6):
    total = total + value
print(total)
```

Values 2,3,4,5 give totals 2,5,9,14. Range(2,2) is empty, so total remains 0. The upper endpoint is excluded.

Judge trace and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the range

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

2, 3 and 4; the stop value is excluded. The body runs three times.

W2 Trace before running

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

W3 Repair the reset

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

Faulty after-addition values are 1, 2, 3 because each starts from zero. Remove the indented total = 0. Correct after-addition values are 1, 3, 6; final **output** 6.

W5 Check a contract limitation

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

No. The written contract requires a nonnegative integer. The script assumes this and does not reject negative values. Empty range behavior does not validate the **input**; report the limitation or add an explicit check.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

W1 Explain the range

W2 Trace before running

W3 Repair the reset

W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

Independent assessment and fresh retry

W4 must preserve the initial 5, update by 2, trace three **iteration**s and print once. Require **output**s 5, 7, 11 for zero one and three rounds.

After support, retry with an initial 4 and additions of 3 for zero one and two rounds: **output**s 4, 7, 10. Record assistance before judging independence.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Keep the portfolio, support and delayed check

> Track code execution separately from a correct paper trace.

Project evidence

Add a labelled **loop**s and **trace table**s **evidence** sheet to Python readiness portfolio. Show how you can trace variable values before and after each **loop iteration**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Support and selected assessment guidance

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

Selected independent answer: Values 2,3,4,5 give totals 2,5,9,14. Range(2,2) is empty, so total remains 0. The upper endpoint is excluded. Keep earlier W4 transfer as additional practice; do not substitute its numbers or key for this selected task. Record prior exposure and use a fresh parallel case after support.

Delayed return

Suggested return points in this level: python-bridge/7, python-bridge/9, python-bridge/12. Use actual classroom dates, adjusting for holidays and gaps.

One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to the next topic

Week 7 stores several values in a list and distinguishes an index from the value at that position.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the teaching cycle.

Planning and assessment reference

Teaching target: Trace variable values before and after each **loop iteration**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Check every learner before discussion: Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: List the supplied values before tracing; range(1, 4) excludes 4.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace range(1,5) with range(1,1). Require a **prediction** and a reason before comparing. Expected result: No values are visited and the final **output** is 0.

Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep individual first attempts separate from supported rehearsal.

Feedback and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

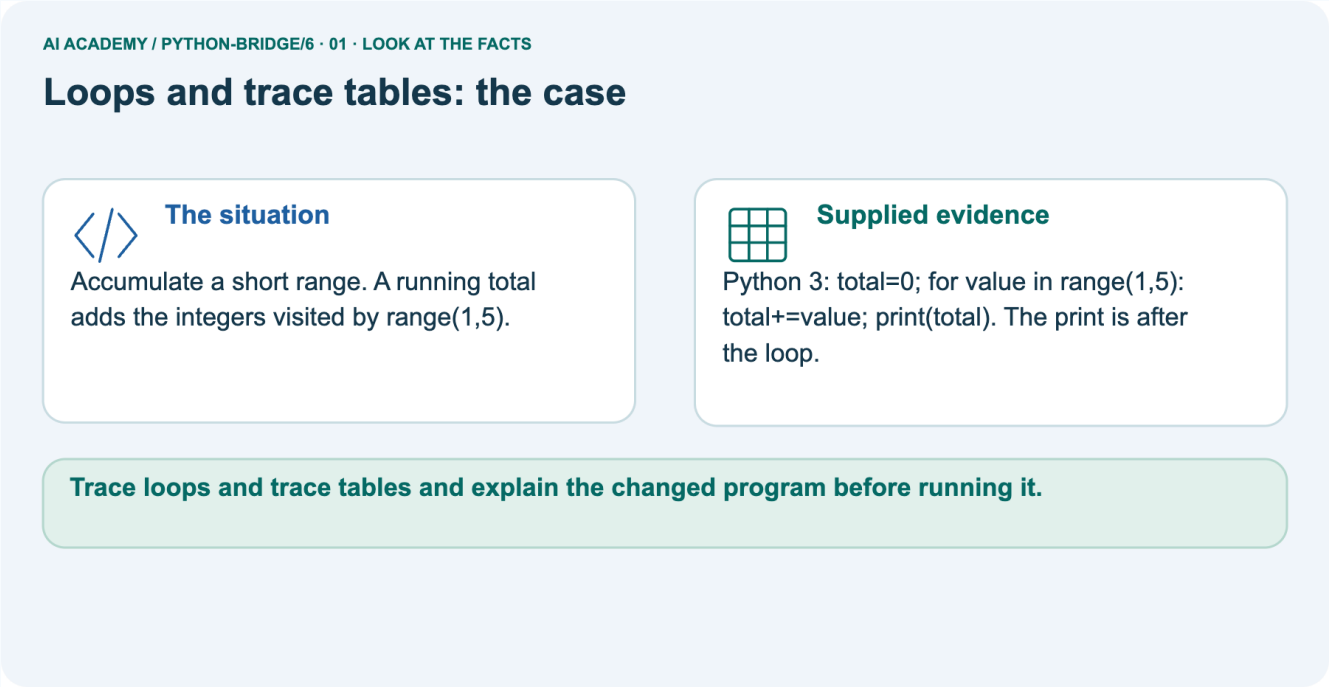
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Use the relevant picture for a specific reasoning move.

01. Loops and trace tables: the case



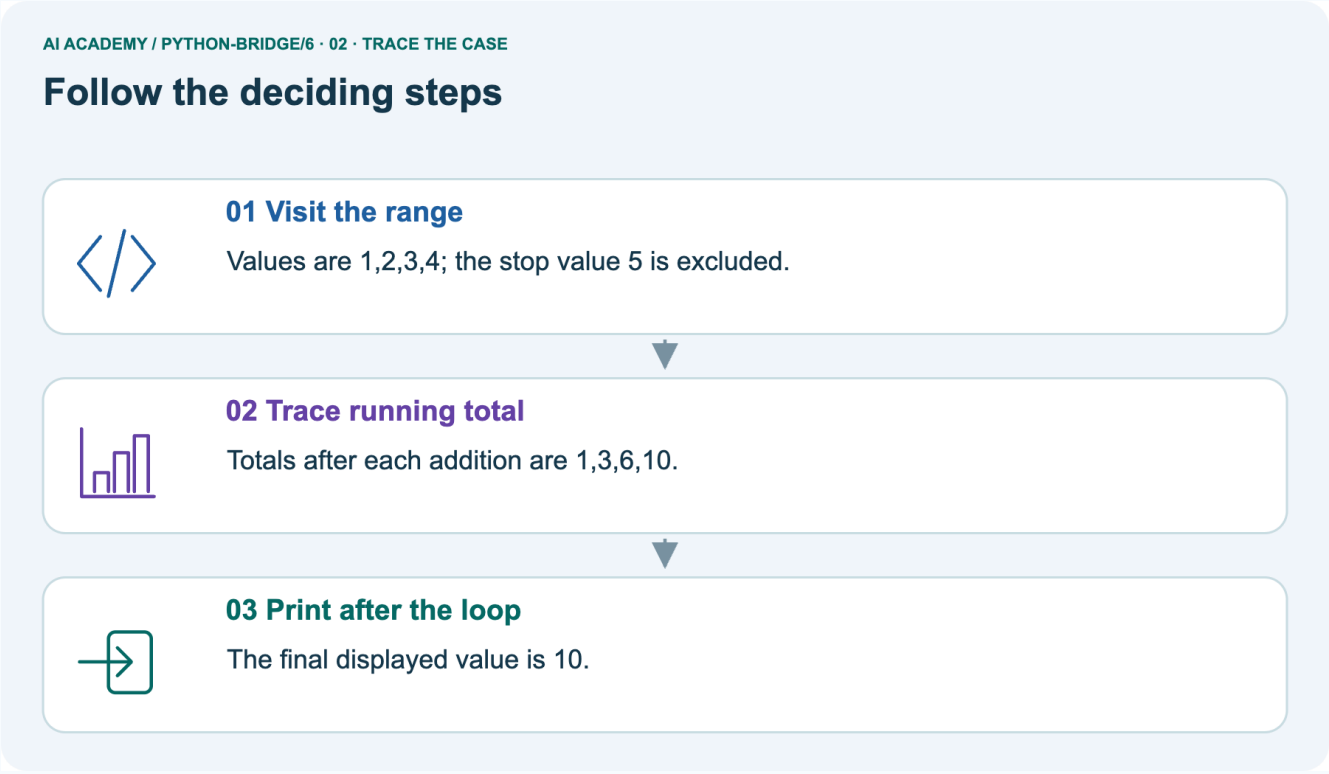
Follow the labelled picture with the matching explanation. Open the vector diagram

Accumulate a short range. A running total adds the integers visited by range(1,5). Supplied facts: Python 3: total=0; for value in range(1,5): total+=value; print(total). The print is after the **loop**.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



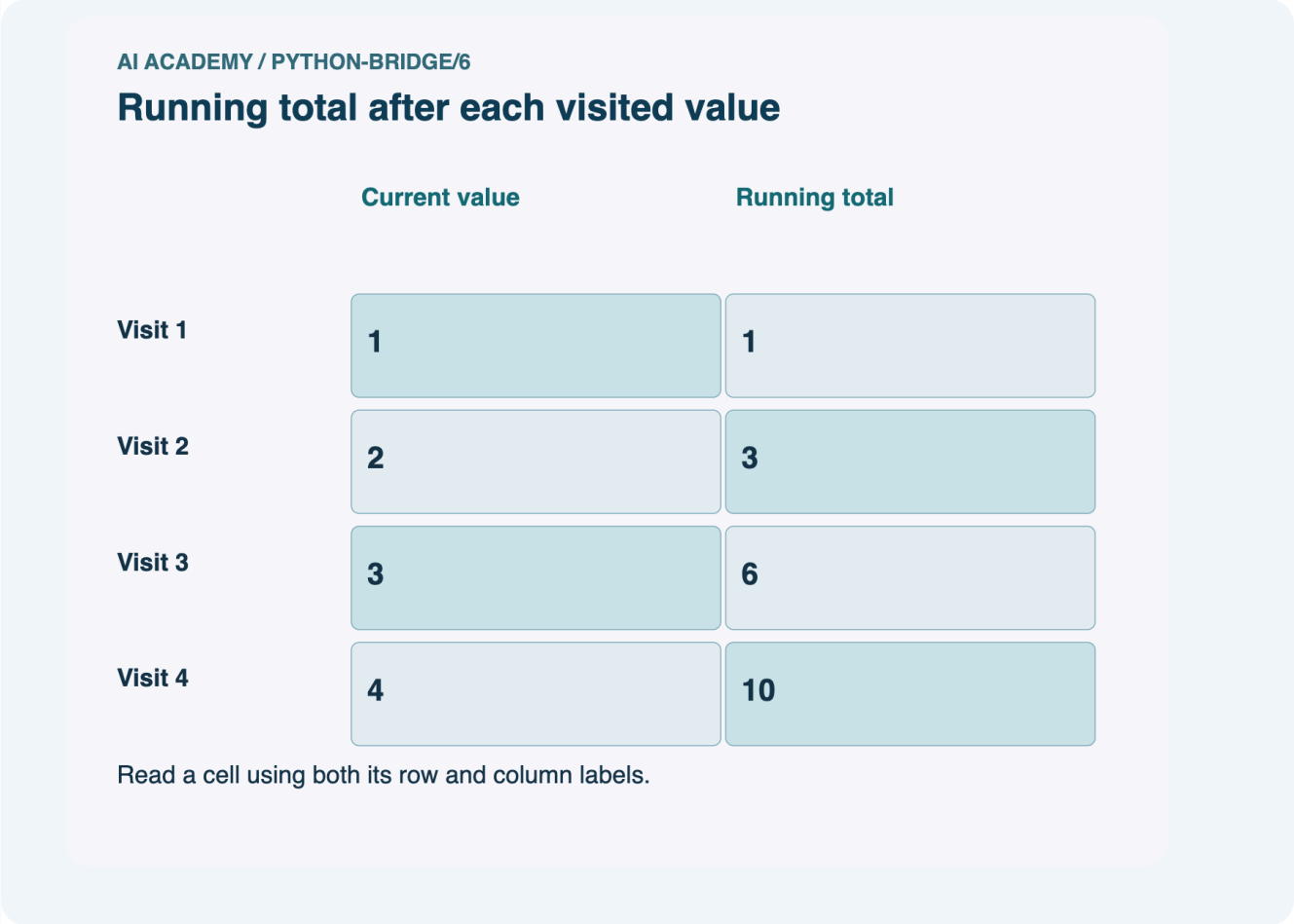
Follow the labelled picture with the matching explanation. Open the vector diagram

Visit the range: Values are 1,2,3,4; the stop value 5 is excluded. → Trace running total: Totals after each addition are 1,3,6,10. → Print after the **loop**: The final displayed value is 10.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Running total after each visited value



Follow the labelled picture with the matching explanation. Open the vector diagram

Running total after each visited value. Visit 1: Current value = 1, Running total = 1. Visit 2: Current value = 2, Running total = 3. Visit 3: Current value = 3, Running total = 6.

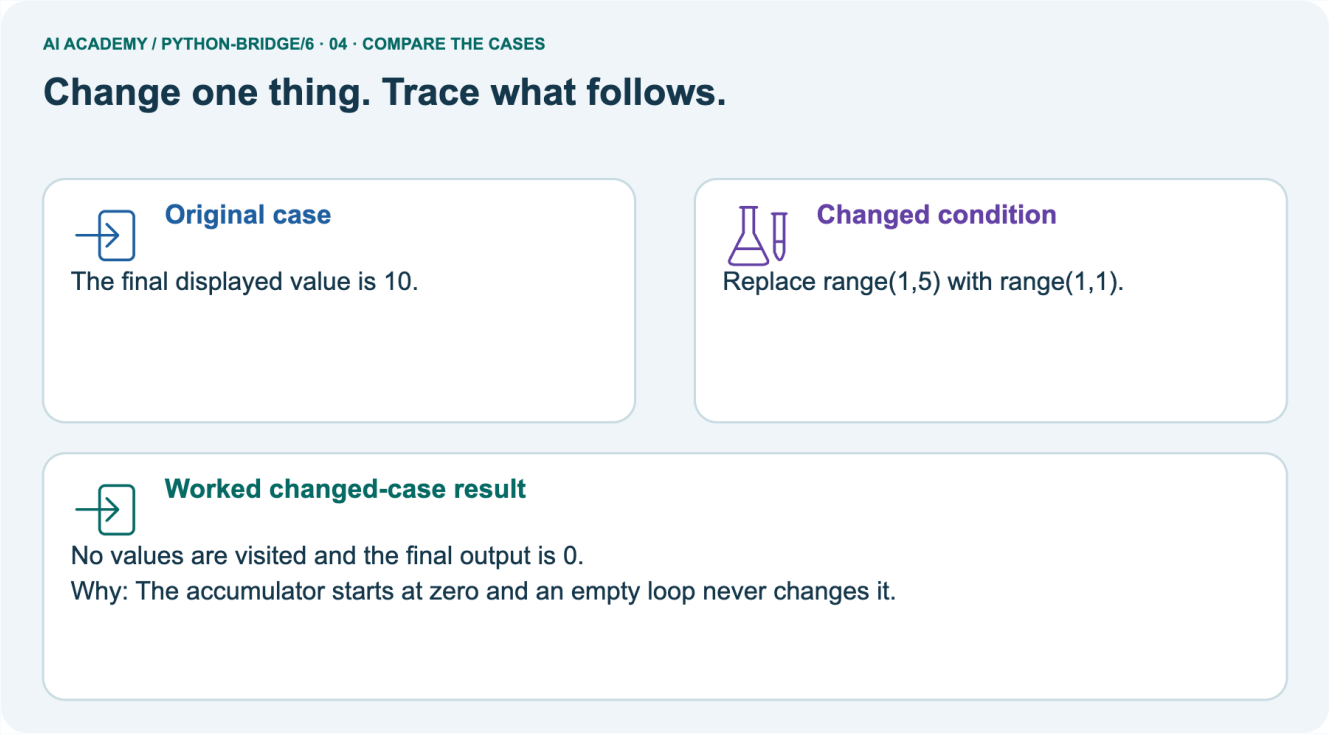
Visit 4: Current value = 4, Running total = 10.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final displayed value is 10. Changed condition: Replace range(1,5) with range(1,1). Expected result: No values are visited and the final **output** is 0.

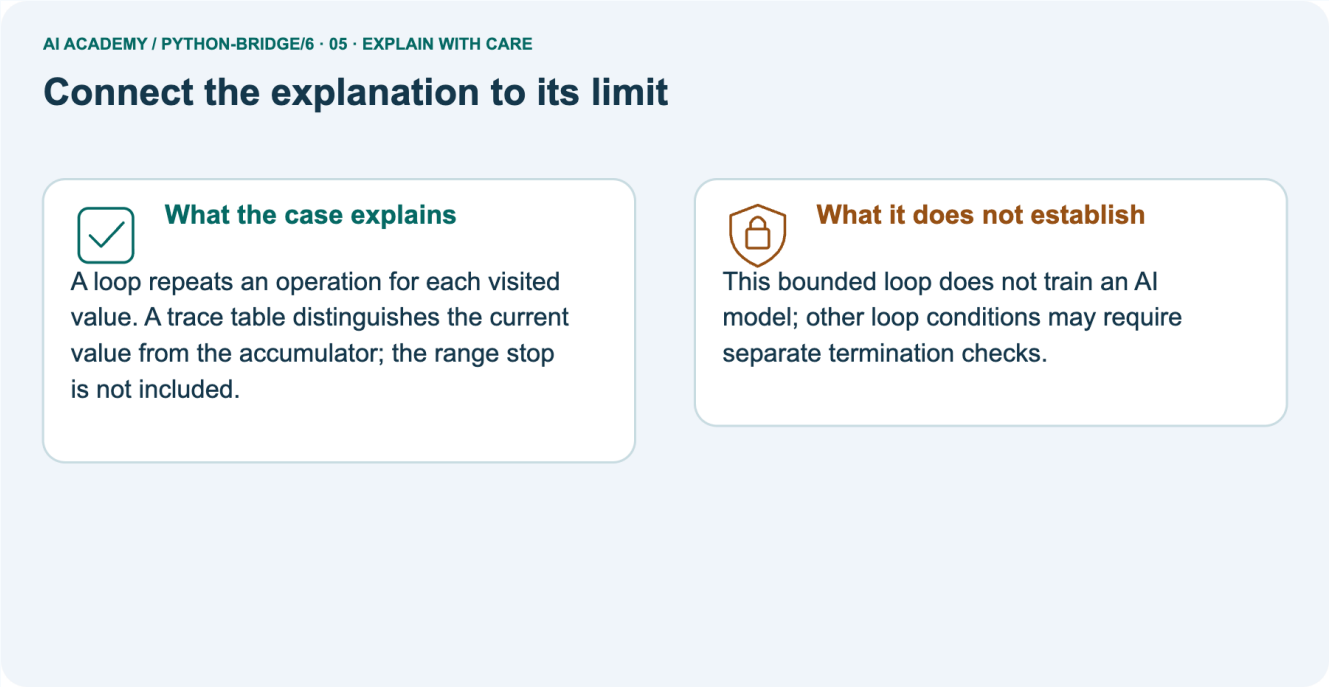
Reason: The **accumulator** starts at zero and an empty **loop** never changes it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **accumulator** starts at zero and an empty **loop** never changes it.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **loop** repeats an operation for each visited value. A **trace table** distinguishes the current value from the **accumulator**; the range stop is not included.

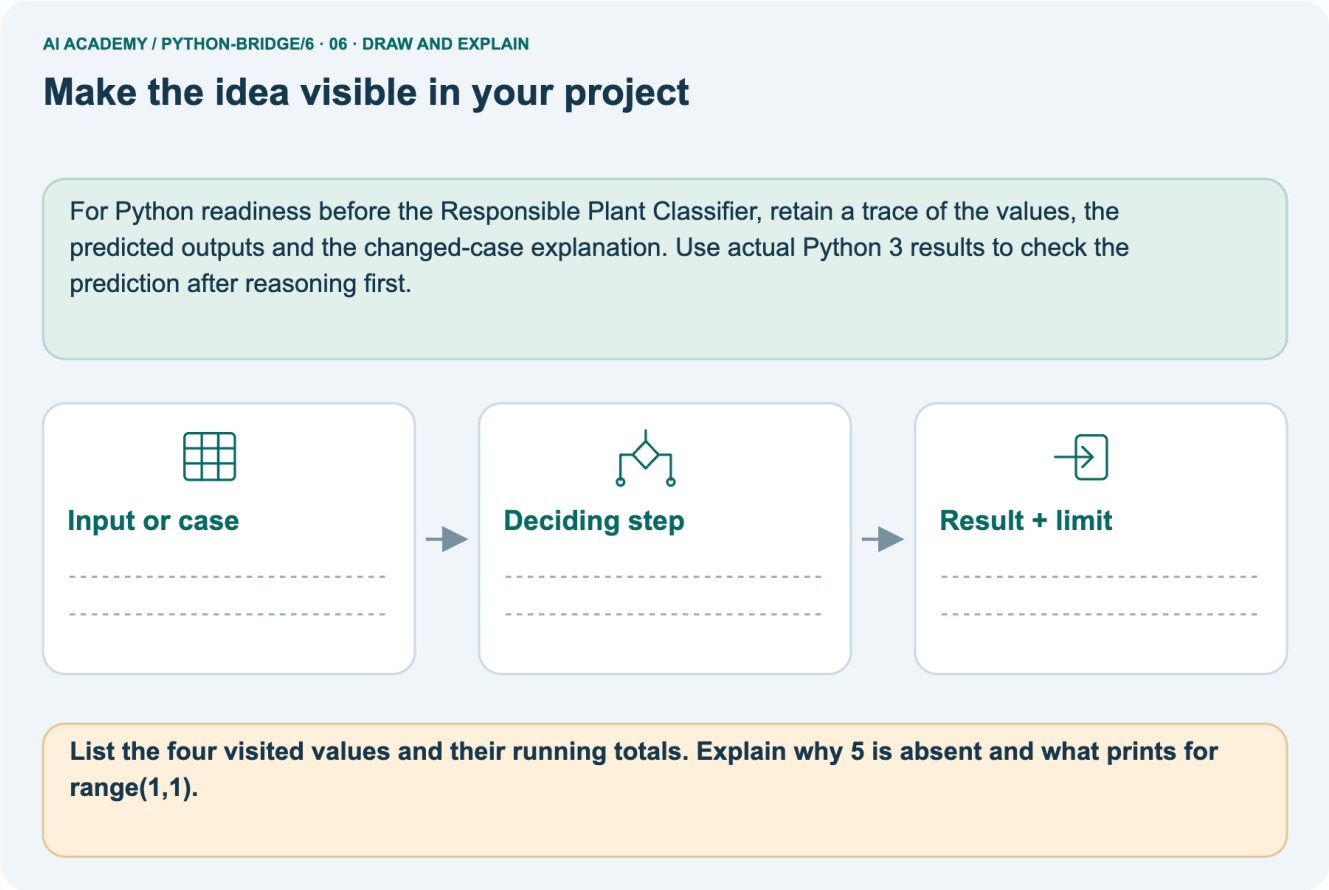
Limit: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This bounded **loop** does not train an AI model; other **loop** conditions may require separate termination checks.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: List the four visited values and their running totals. Explain why 5 is absent and what prints for range(1,1).

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The visits are 1,2,3,4 and totals are 1,3,6,10. The stop is exclusive. An empty range leaves total at its initial 0.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Loops and trace tables: the case

02. Follow the deciding steps

03. Running total after each visited value

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.